# CLOUD COMPUTING

# UNIT-3

## Infrastructure
## &
## Advanced Techniques In Cloud Computing

# Infrastructure

- **Existing cloud infrastructure**
- The cloud computing infrastructure at Amazon, Google, and Microsoft (as of mid 2012).
  - Amazon is a pioneer in Infrastructure-as-a-Service (IaaS).
  - Google's efforts are focused on Software-as-a-Service (SaaS) and Platform-as-a-Service (PaaS).
  - Microsoft is involved in PaaS.
- Private clouds are an alternative to public clouds. Open-source cloud computing platforms such as:
  - Eucalyptus,
  - OpenNebula,
  - Nimbus,
  - OpenStack
- can be used as a control infrastructure for a private cloud.

# Infrastructure

- **Eucalyptus** (Elastic Utility Computing Architecture for Linking Your Programs to Useful Systems) is an open-source software platform that enables organizations to build private and hybrid cloud environments
- **OpenNebula** is an open-source cloud computing platform that focuses on simplifying the management of heterogeneous data center infrastructures (Ex, file & database servers).
- **Nimbus** is an open-source toolkit tailored for creating and managing cloud infrastructure. It is specifically designed for scientific and research communities.
- **OpenStack** is a widely used open-source cloud computing platform that provides a services for building and managing public, private, and hybrid clouds can be used as a control infrastructure for a private cloud

# Cloud Computing at Amazon

- **Amazon Web Services (AWS)**
  - IaaS cloud computing services launched in 2006 by AWS.
  - Businesses in 200+ countries used AWS
  - The infrastructure consists of compute and storage servers interconnected by high-speed networks and supports a set of services.
  - An application developer: Installs applications on a platform of his/her choice Manages resources allocated by Amazon.
  - Amazon was the first provider of cloud computing, which release of its Elastic Computing platform called EC2
  - Elastic Compute Cloud (EC2) is a Web service with a simple interface for launching instances of an application under several operating systems, such as Microsoft Windows Server.
  - EC2 Is based on the Xen virtualization strategy
  - Simple Storage System (S3) is a storage service designed to store large objects. It supports a minimal set of functions: write, read, and delete.

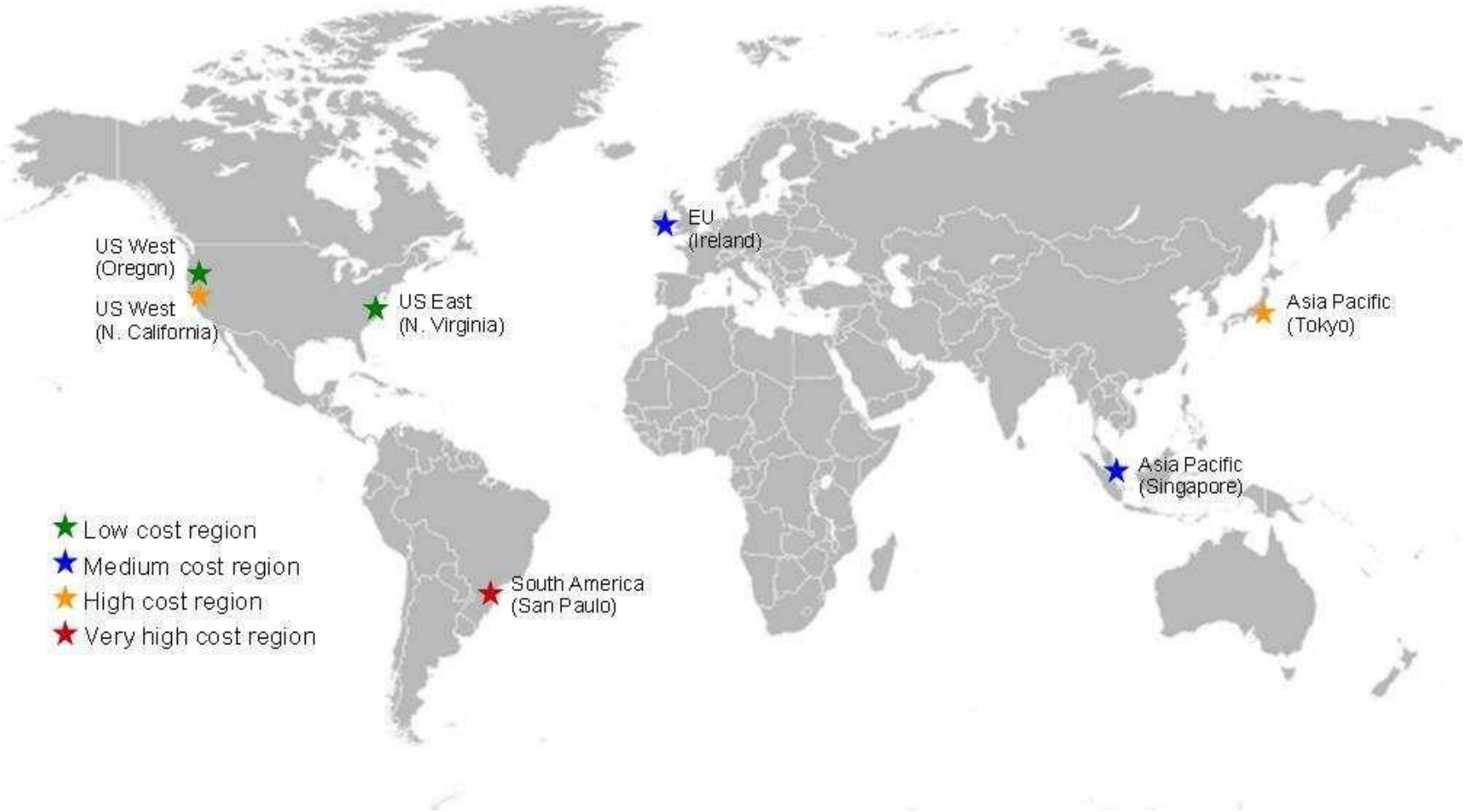
# AWS regions and availability zones

- Amazon offers cloud services through a network of data centers on several continents.
- In each *region* there are several availability zones interconnected by high-speed networks.
- An *availability zone* is a data center consisting of a large number of servers.

| Region | Location | Availability zones | Cost |
|---|---|---|---|
| US West | Oregon | us-west-2a/2b/2c | Low |
| US West | North California | us-west-1a/1b/1c | High |
| US East | North Virginia | us-east-1a/2a/3a/4a | Low |
| Europe | Ireland | eu-west-1a/1b/1c | Medium |
| South America | Sao Paulo, Brazil | sa-east-1a/1b | Very high |
| Asia Pacific | Tokyo, Japan | ap-northeast-1a/1b | High |
| Asia Pacific | Singapore | ap-southeast-1a/1b | Medium |

- Regions do not share resources and communicate through the Internet.

# AWS regions and availability zones



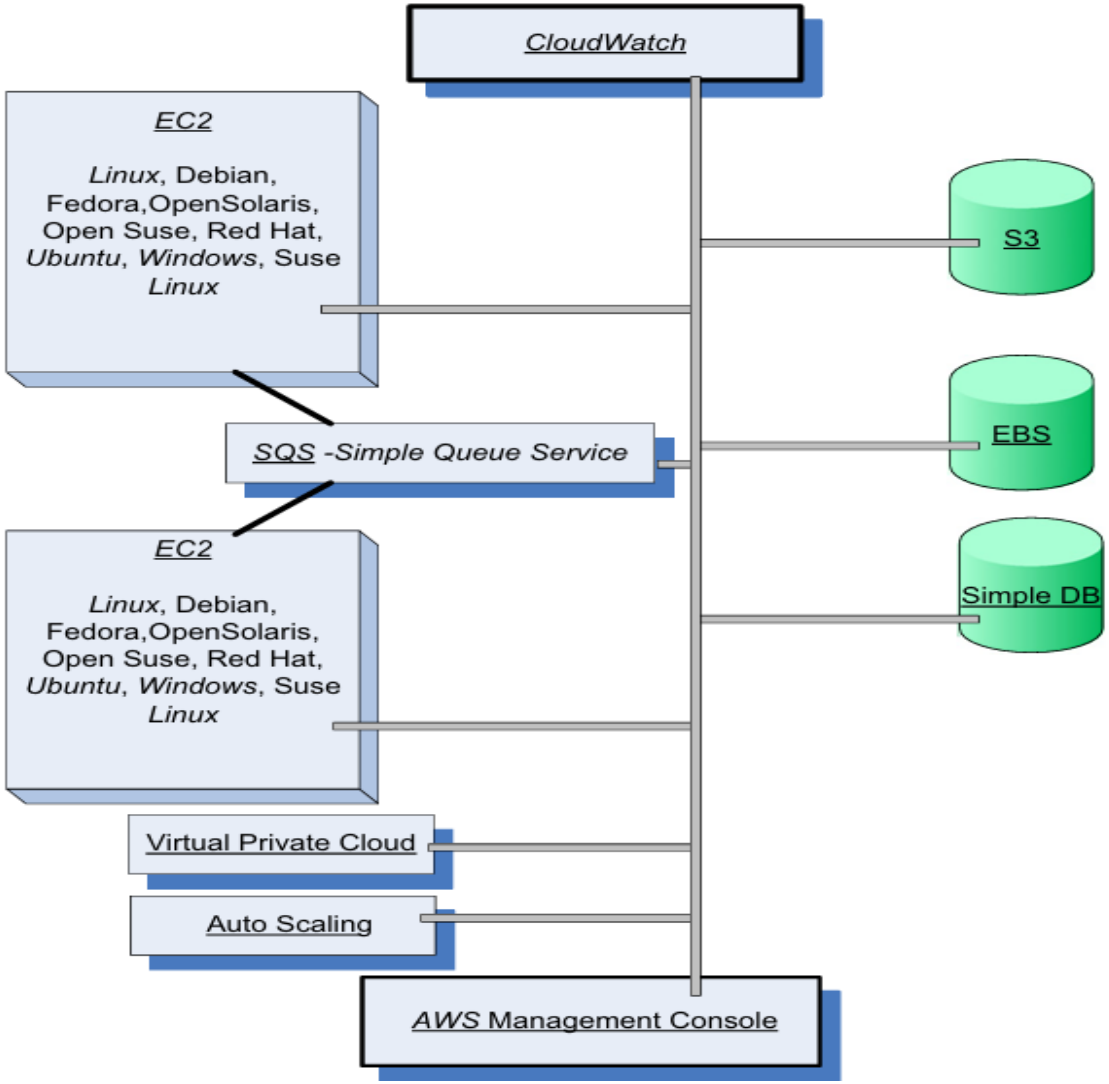
# AWS instances

- An instance is a virtual server with a well specified set of resources including: CPU cycles, main memory, secondary storage, communication and I/O bandwidth.
- The user chooses:
  - The region and the availability zone where this virtual server should be placed.
  - An instance type from a limited menu of instance types.
- When launched, an instance is provided with a DNS name; this name maps to a
  - *private IP address* → for internal communication within the internal EC2 communication network.
  - *public IP address* → for communication outside the internal Amazon network, e.g., for communication with the user that launched the instance.
- Network Address Translation (NAT) maps external IP addresses to internal ones.

- The public IP address is assigned for the lifetime of an instance.
- An instance can request an *elastic IP address*, rather than a public IP address. The elastic IP address is a static public IP address allocated to an instance from the available pool of the availability zone.
- An elastic IP address is not released when the instance is stopped or terminated and must be released when no longer needed.

# Examples of Amazon Web Services

- *AWS Management Console* - allows users to access the services offered by AWS .
- *Elastic Cloud Computing (EC2)* - allows a user to launch a variety of operating systems.
- *Simple Queuing Service (SQS)* - allows multiple *EC2* instances to communicate with one another.
- *Simple Storage Service (S3)*, *Simple DB*, and *Elastic Bloc Storage (EBS)* - storage services.
- *Cloud Watch* - supports performance monitoring.
- *Auto Scaling* - supports elastic resource management.
- *Virtual Private Cloud* - allows direct migration of parallel applications.

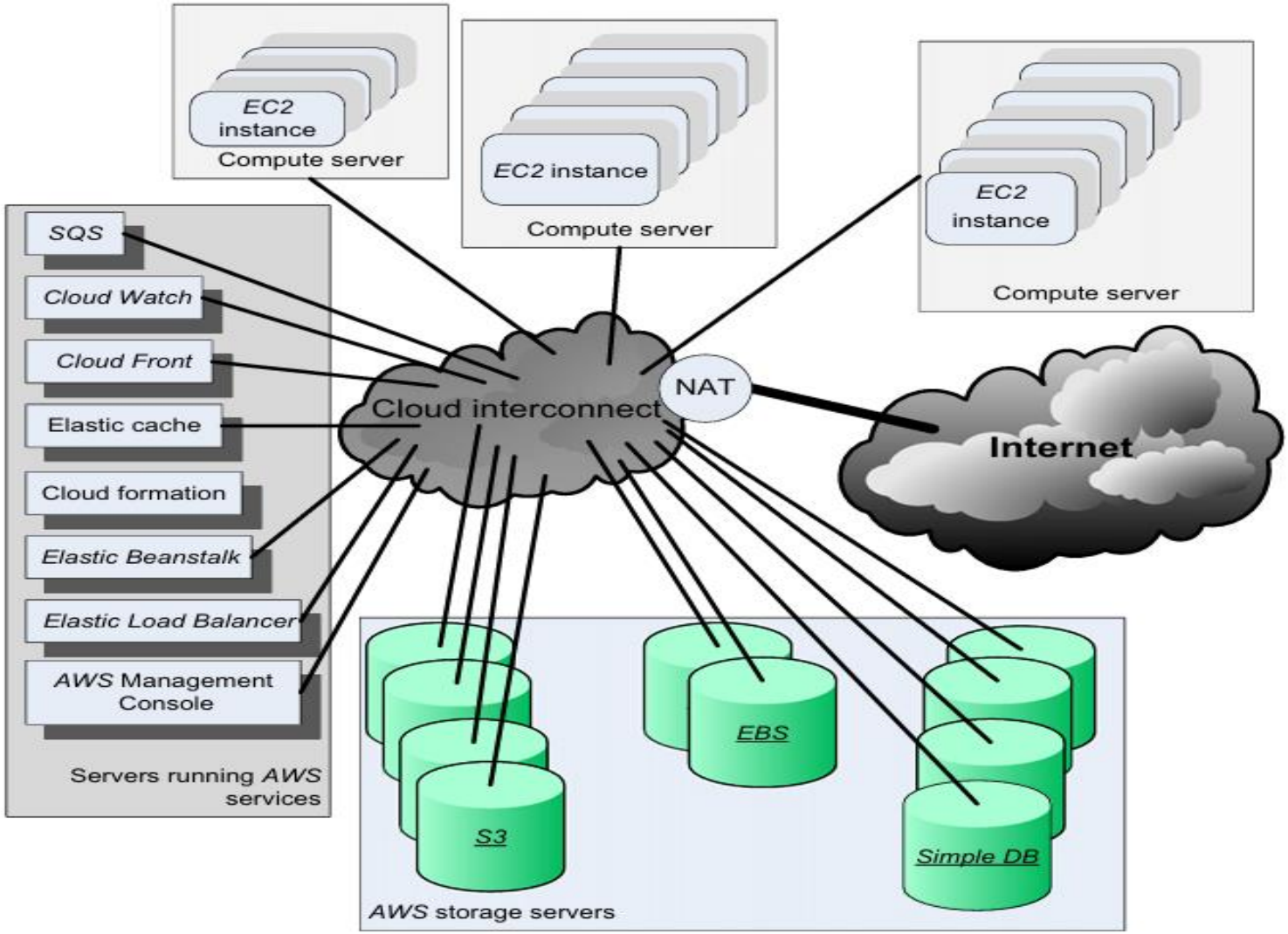

CloudWatch
EC2
Linux, Debian, Fedora,OpenSolaris, Open Suse, Red Hat, Ubuntu, Windows, Suse Linux
S3
EBS
SQS -Simple Queue Service
EC2
Linux, Debian, Fedora,OpenSolaris, Open Suse, Red Hat, Ubuntu, Windows, Suse Linux
Simple DB
Virtual Private Cloud
Auto Scaling
AWS Management Console

- *EC2* - web service for launching instances of an application under several operating systems, such as:
  - Several Linux distributions.
  - Microsoft Windows Server 2003 and 2008.
  - OpenSolaris.
  - FreeBSD.
  - NetBSD.
- A user can
  - Load an *EC2* instance with a custom application environment.
  - Manage network's access permissions.
  - Run the image using as many or as few systems as desired
- Import virtual machine (VM) images from the user environment to an instance through VM import.
- EC2 instances boot from an AMI (Amazon Machine Image) digitally
- signed and stored in S3.

- Users can access:
  - Images provided by Amazon.
  - Customize an image and store it in S3.
- An EC2 instance is characterized by the resources it provides:
  - VC (Virtual Computers) – virtual systems running the instance.
  - CU (Compute Units) – measure computing power of each system.
  - Memory.
  - I/O capabilities.
- It also automatically distributes the incoming application traffic among multiple instances using the elastic load-balancing facility
- A user can interact with EC2 using a set of SOAP (Simple Object Access Protocol) messages (It is a messaging protocol that allows programs running on different operating systems and platforms (e.g., Windows, Linux) to communicate over a network)

- Instance types
  - Standard instances: micro (StdM), small (StdS), large (StdL), extra large (StdXL); small is the default.
  - High memory instances: high-memory extra large (HmXL), high-memory double extra large (Hm2XL), and high-memory quadruple extra large (Hm4XL).
  - High CPU instances: high-CPU extra large (HcpuXL).
  - Cluster computing: cluster computing quadruple extra large (Cl4XL).

| Instance name | API name | Platform (32/64-bit) | Memory (GB) | Max EC2 compute units | I-memory (GB) | I/O (M/H) |
|---|---|---|---|---|---|---|
| StdM | | 32 and 64 | 0. 633 | 1 VC; 2 CUs | | |
| StdS | `m1.small` | 32 | 1.7 | 1 VC; 1 CU | 160 | M |
| StdL | `m1.large` | 64 | 7.5 | 2 VCs; $2 \times 2$ CUs | 85 | H |
| StdXL | `m1.xlarge` | 64 | 15 | 4 VCs; $4 \times 2$ CUs | 1,690 | H |
| HmXL | `m2.xlarge` | 64 | 17.1 | 2 VCs; $2 \times 3.25$ CUs | 420 | M |
| Hm2XL | `m2.2xlarge` | 64 | 34.2 | 4 VCs; $4 \times 3.25$ CUs | 850 | H |
| Hm4XL | `m2.4xlarge` | 64 | 68.4 | 8 VCs; $8 \times 3.25$ CUs | 1,690 | H |
| HcpuXL | `c1.xlarge` | 64 | 7 | 8 VCs; $8 \times 2.5$ CUs | 1,690 | H |
| Cl4XL | `cc1.4xlarge` | 64 | 18 | 33.5 CUs | 1,690 | H |


EC2 instance
Compute server
EC2 instance
Compute server
EC2 instance
Compute server
SQS
Cloud Watch
Cloud Front
Elastic cache
Cloud formation
Elastic Beanstalk
Elastic Load Balancer
AWS Management Console
Servers running AWS services
Cloud interconnect
NAT
Internet
S3
EBS
Simple DB
AWS storage servers

# Simple Storage System

- Service designed to store large objects; an application can handle an unlimited number of objects ranging in size from 1 byte to 5 TB.
- An object is stored in a bucket and retrieved via a unique, developer - assigned key; a bucket can be stored in a Region selected by the user.
- Supports a minimal set of functions: write, read, and delete; it does not support primitives to copy, to rename, or to move an object from one bucket to another.
- The object names are global.
- S3 maintains for each object: the name, modification time, an access control list, and up to 4 KB of user-defined metadata.
  - Authentication mechanisms ensure that data is kept secure.
  - Object Storage: S3 stores data as objects

# Simple Storage System

- Objects can be made public, and rights can be granted to other users.
- *S3* computes the MD5 of every object written and returns it in a field called ETag.
- A user is expected to compute the MD5 of an object stored or written and compare this with the ETag; if the two values do not match, then the object was corrupted during transmission or storage.

# Elastic Block Store (EBS)

"A virtual hard disk that you can attach, detach, and reattach to your EC2 instance whenever you want — and your data stays safe."

- Provides persistent block level storage volumes for use with *EC2* instances; suitable for database applications, file systems, and applications using raw data devices.
- A volume appears to an application as a raw, unformatted and reliable physical disk; the range 1 GB -1 TB.
- An *EC2* instance may mount multiple volumes, but a volume cannot be shared among multiple instances.
- EBS supports the creation of snapshots of the volumes attached to an instance and then uses them to restart the instance.
- The volumes are grouped together in Availability Zones and are automatically replicated in each zone.

## ⚙️ Key Features of EBS

| Feature | Description |
|---|---|
| **Persistent Storage** | Data remains available even after instance termination (if not deleted). |
| **Scalable** | You can increase volume size, change type, or adjust performance without downtime. |
| **Highly Available** | Data is automatically replicated within an Availability Zone (AZ). |
| **Backup & Snapshots** | You can create EBS **snapshots** (point-in-time backups) stored in Amazon S3. |
| **Performance Options** | Offers different volume types optimized for throughput or IOPS (performance). |
| **Encryption** | Supports **encryption at rest** and **in transit** using AWS KMS keys. |

# SimpleDB

Amazon SimpleDB is a simple, flexible, NoSQL database for storing and querying small structured datasets — without worrying about database management or servers.

- Non-relational data store. Supports store and query functions traditionally provided only by relational databases.
- Supports high performance Web applications; users can store and query data items via Web services requests.
- Creates multiple geographically distributed copies of each data item.
- It manages automatically:
    - The infrastructure provisioning.
    - Hardware and software maintenance.
    - Replication and indexing of data items.
    - Performance tuning

## ⚙️ Key Features

| Feature | Description |
|---|---|
| **Serverless** | No need to manage servers, hardware, or configurations. |
| **Schema-less** | You can store data without defining a fixed table structure. |
| **Automatic Indexing** | Automatically indexes all attributes for quick searching. |
| **Scalable** | Handles growing data and traffic automatically. |
| **Integrated with AWS** | Works with EC2 and S3 for web app development. |
| **Pay-per-use** | You pay only for the storage and data transfer you use. |

# SQS-Simple Queue Service

Amazon SQS is like a message postbox between systems - one service drops a message (task), another picks it up later and processes it, ensuring smooth, scalable, and reliable communication.

- Hosted message queues are accessed through standard SOAP and Query interfaces.
- Supports automated workflows - *EC2* instances can coordinate by sending and receiving SQS messages.
- Applications using SQS can run independently and asynchronously, and do not need to be developed with the same technologies.
- A received message is “locked" during processing; if processing fails, the lock expires and the message is available again.
- Queue sharing can be restricted by IP address and time-of-day.

## ⚙️ Key Features of SQS

| Feature | Description |
|---|---|
| **Fully Managed** | No need to manage servers or message brokers. |
| **Scalable** | Automatically handles millions of messages per second. |
| **Asynchronous Communication** | Producers and consumers work independently. |
| **Durable Storage** | Messages are redundantly stored across multiple servers. |
| **Security** | Supports encryption (SSE), IAM policies, and VPC endpoints. |
| **Visibility Timeout** | Prevents multiple consumers from processing the same message. |
| **Dead Letter Queues (DLQ)** | Stores failed messages for later debugging. |

- Meaning of Amazon SQS Amazon Simple Queue Service (SQS) allows different parts of an application (like microservices, distributed systems, or serverless functions) to communicate by sending, storing, and receiving messages in a queue - without needing to connect directly or run at the same time.
- It helps ensure reliable, scalable, and fault-tolerant message delivery between producers and consumers.

➢ How It Works
  - Producer (Sender) — sends a message to the SQS queue.
  - Queue — stores the message safely until it’s processed.
  - Consumer (Receiver) — retrieves and processes the message.
  - Once processed, the message is deleted from the queue.

## 🧠 Important SQS Concepts

| Concept | Description |
|---|---|
| **Message Retention Period** | How long a message stays in the queue (default 4 days, up to 14 days). |
| **Visibility Timeout** | Time during which a message is hidden from other consumers after being fetched. |
| **Polling** | The process of retrieving messages (Short or Long polling). |
| **Delay Queue** | Delays message delivery by a specific time (up to 15 minutes). |
| **Dead Letter Queue (DLQ)** | Stores messages that couldn't be processed successfully after several attempts. |

# CloudWatch

- Monitoring infrastructure used by application developers, users, and system administrators to collect and track metrics important for optimizing the performance of applications and for increasing the efficiency of resource utilization.
- Without installing any software a user can monitor either seven or eight pre-selected metrics and then view graphs and statistics for these metrics.
- When launching an Amazon Machine Image (AMI) the user can start the CloudWatch and specify the type of monitoring:
  - Basic Monitoring - free of charge; collects data at five-minute intervals for up to seven metrics.
  - Detailed Monitoring - subject to charge; collects data at one minute interval.

# AWS services introduced in 2012

- *Route 53* - low-latency DNS service used to manage user's DNS public records.
- *Elastic MapReduce (EMR)* - supports processing of large amounts of data using a hosted Hadoop running on *EC2*.
- *Simple Workflow Service (SWF)* - supports workflow management; allows scheduling, management of dependencies, and coordination of multiple *EC2* instances.
- *ElastiCache* - enables web applications to retrieve data from a managed in-memory caching system rather than a much slower disk-based database.
- *DynamoDB* - scalable and low-latency fully managed NoSQL database service.
- *CloudFront* - web service for content delivery.
- *Elastic Load Balancer* - automatically distributes the incoming requests across multiple instances of the application.

- *Elastic Beanstalk* - handles automatically deployment, capacity provisioning, load balancing, auto-scaling, and application monitoring functions.
- *CloudFormation* - allows the creation of a stack describing the infrastructure for an application.

## Elastic Beanstalk

- Handles automatically the deployment, capacity provisioning, load balancing, auto-scaling, and monitoring functions.
- Interacts with other services including *EC2*, *S3*, SNS, Elastic Load Balance and AutoScaling.
- The management functions provided by the service are:
    - Deploy a new application version (or rollback to a previous version).
    - Access to the results reported by CloudWatch monitoring service.
    - Email notifications when application status changes or application servers are added or removed.
    - Access to server log files without needing to login to the application servers.
- The service is available using: a Java platform, the PHP server-side description language, or the .NET framework

# SaaS services offered by Google

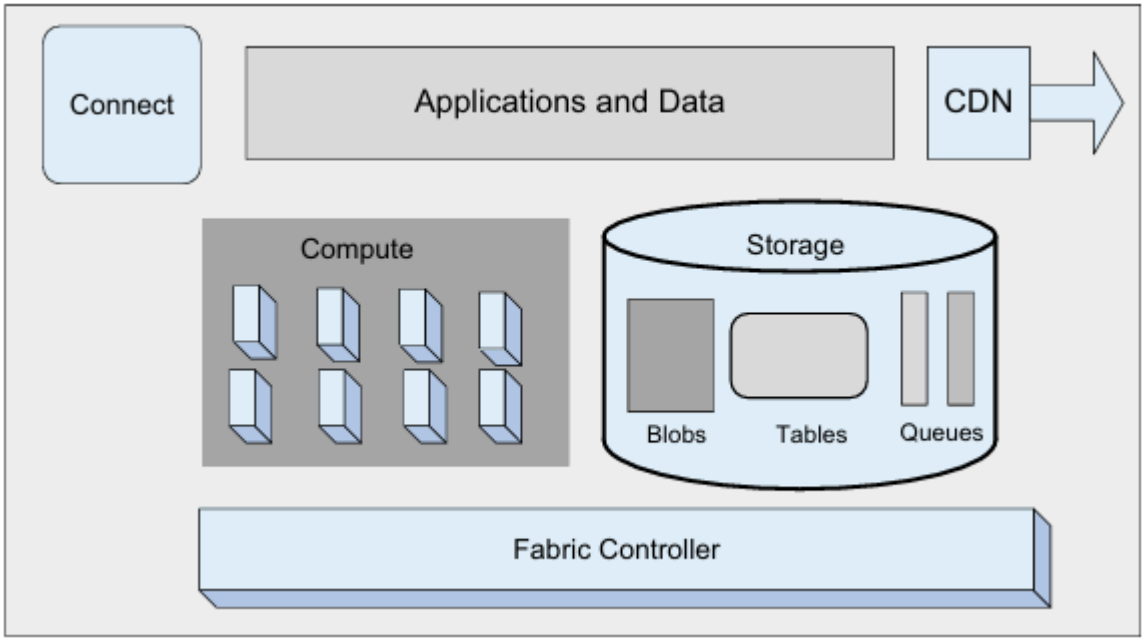
- *Gmail* - hosts Emails on Google servers and provides a web interface to access the Email.
- *Google docs* - a web-based software for building text documents, spreadsheets and presentations.
- *Google Calendar* - a browser-based scheduler; supports multiple user calendars, calendar sharing, event search, display of daily/weekly/monthly views, and so on.
- *Google Groups* - allows users to host discussion forums to create messages online or via Email.
- *Picasa* - a tool to upload, share, and edit images.
- *Google Maps* - web mapping service; offers street maps, a route planner, and an urban business locator for numerous countries around the world

# PaaS services offered by Google

- *AppEngine* - a developer platform hosted on the cloud.
  - Initially supported Python, Java was added later.
  - The database for code development can be accessed with GQL (Google Query Language) with a SQL-like syntax.
- *Google Co-op* - allows users to create customized search engines based on a set of facets/categories.
- *Google Drive* - an online service for data storage.
- *Google Base* - allows users to load structured data from different sources to a central repository, a very large, self-describing, semi-structured, heterogeneous database.

# PaaS and SaaS services from Microsoft

- *Windows Azure* - an operating system; has 3 components:
  - Compute - provides a computation environment.
  - Storage - for scalable storage.
  - Fabric Controller - deploys, manages, and monitors applications.
- *SQL Azure* - a cloud-based version of the SQL Server.
- *Azure AppFabric*, formerly .NET Services - a collection of services for cloud applications.



# Open-source platforms for private clouds

- *Eucalyptus* - can be regarded as an open-source counterpart of Amazon's EC2.

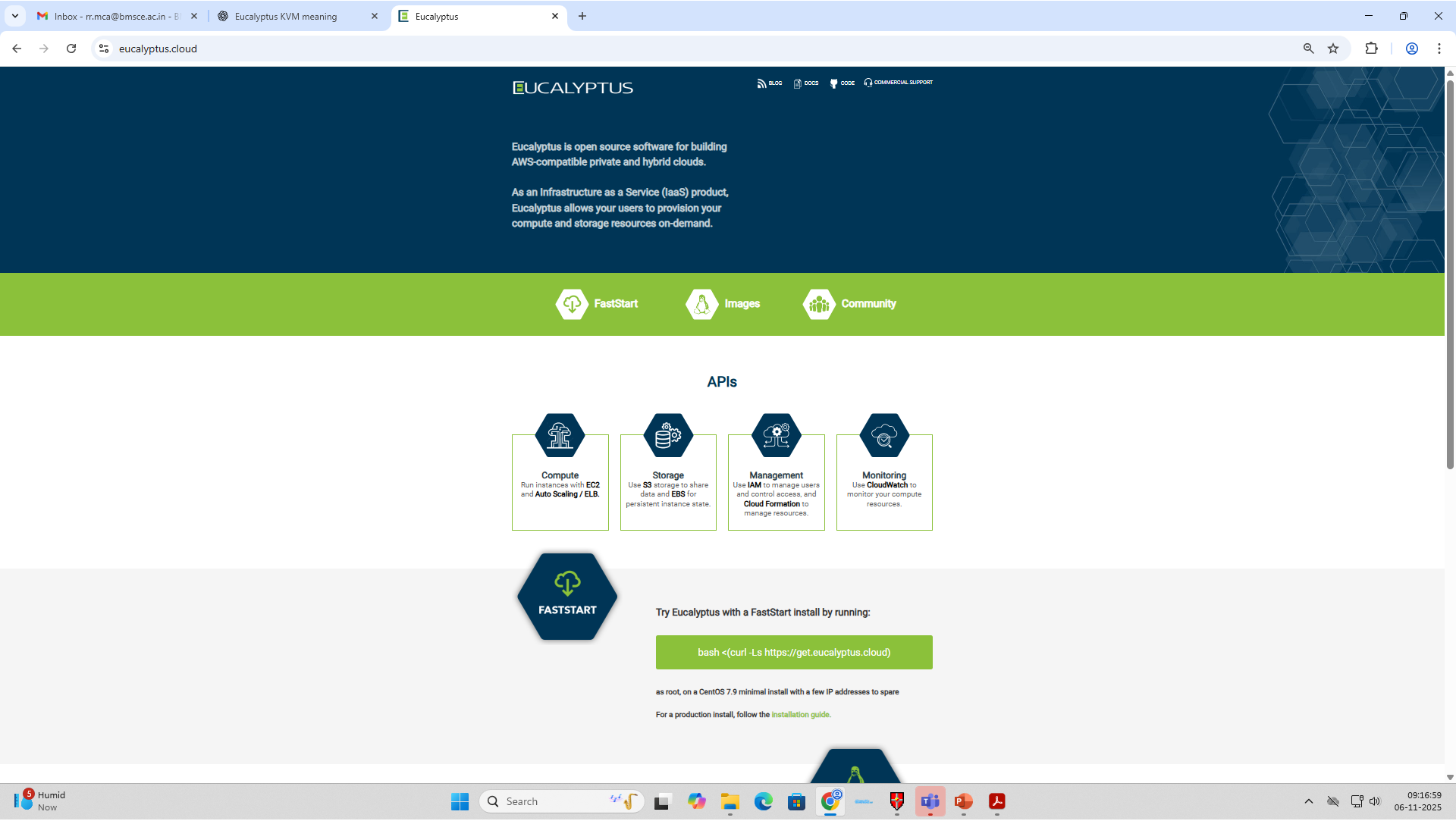
- *Open-Nebula* - a private cloud with users actually logging into the head node to access cloud functions. The system is centralized and its default configuration uses the NFS file system.

- *Nimbus* - a cloud solution for scientific applications based on Globus software, inherits from Globus:
    - The image storage.
    - The credentials for user authentication.
    - The requirement that a running Nimbus process can **ssh** into all compute nodes.

# Eucalyptus

- *Virtual Machines* - run under several VMMs including Xen, KVM, and VMware.
- *Node Controller* - runs on server nodes hosting a VM and controls the activities of the node.
- *Cluster Controller* - controls a number of servers.
- *Cloud Controller* - provides the cloud access to end-users, developers, and administrators.
- *Storage Controller* - provides persistent virtual hard drives to applications. It is the correspondent of EBS.
- *Storage Service (Walrus)* - provides persistent storage; similar to S3, it allows users to store objects in buckets.

# Eucalyptus


Inbox - rr.mca@bmsce.ac.in - B
Eucalyptus KVM meaning
Eucalyptus
eucalyptus.cloud
EUCALYPTUS
BLOG
DOCS
CODE
COMMERCIAL SUPPORT
Eucalyptus is open source software for building AWS-compatible private and hybrid clouds.
As an Infrastructure as a Service (IaaS) product, Eucalyptus allows your users to provision your compute and storage resources on-demand.
FastStart
Images
Community
APIs
Compute
Run instances with EC2 and Auto Scaling / ELB.
Storage
Use S3 storage to share data and EBS for persistent instance state.
Management
Use IAM to manage users and control access, and Cloud Formation to manage resources.
Monitoring
Use CloudWatch to monitor your compute resources.
FASTSTART
Try Eucalyptus with a FastStart install by running:
bash <(curl -Ls https://get.eucalyptus.cloud)
as root, on a CentOS 7.9 minimal install with a few IP addresses to spare
For a production install, follow the installation guide.
Humid
Now
Search
09:16:59
06-11-2025

# Comparision

**Table 3.5** A side-by-side comparison of *Eucalyptus*, *OpenNebula*, and *Nimbus*.

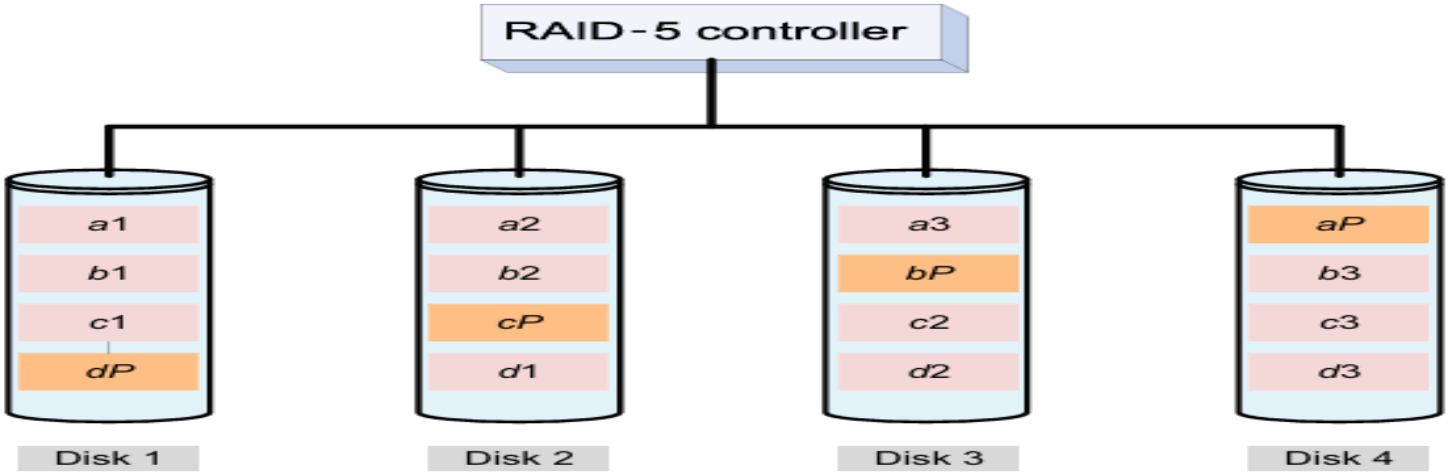
| | ***Eucalyptus*** | ***OpenNebula*** | ***Nimbus*** |
|---|---|---|---|
| Design | Emulate *EC2* | Customizable | Based on Globus |
| Cloud type | Private | Private | Public/Private |
| User population | Large | Small | Large |
| Applications | All | All | Scientific |
| Customizability | Administrators and limited users | Administrators and users | All but image storage and credentials |
| Internal security | Strict | Loose | Strict |
| User access | User credentials | User credentials | x509 credentials |
| Network access | To cluster controller | — | To each compute node |

# Cloud storage diversity and vendor lock-in

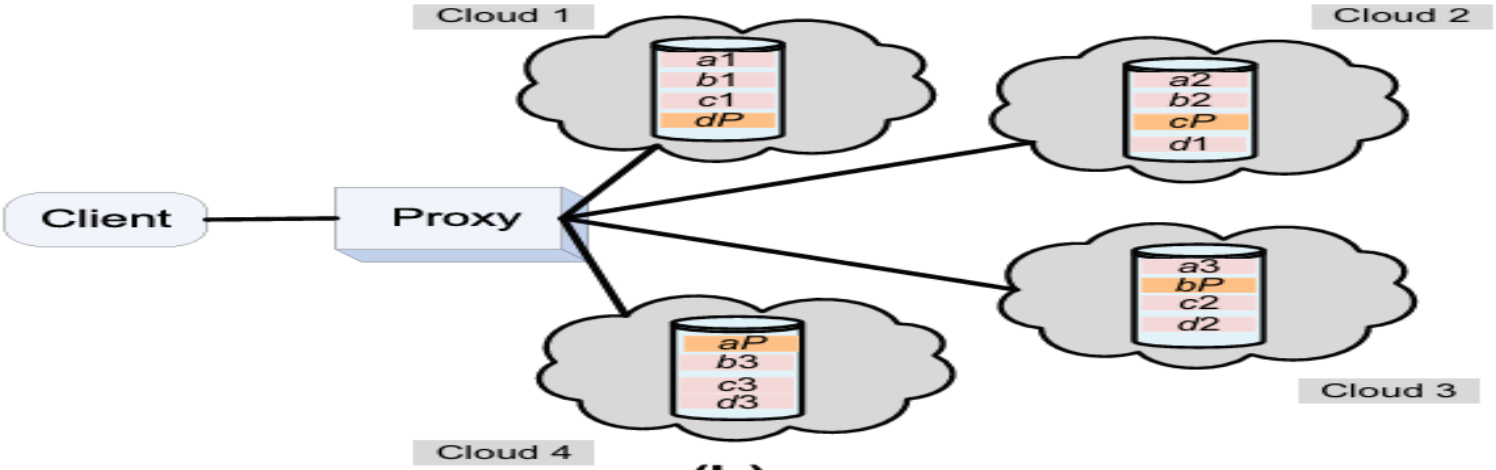
- Risks when a large organization relies on a single cloud service provider:
  - Cloud services may be unavailable for a short or an extended period of time.
  - Permanent data loss in case of a catastrophic system failure.
  - The provider may increase the prices for service.

- Switching to another provider could be very costly due to the large volume of data to be transferred from the old to the new provider.

- A solution is to replicate the data to multiple cloud service providers, similar to data replication in RAID.


RAID-5 controller
a1
b1
c1
dP
Disk 1
a2
b2
cP
d1
Disk 2
a3
bP
c2
d2
Disk 3
aP
b3
c3
d3
Disk 4

(a)


Cloud 1
a1
b1
c1
dP
Cloud 2
a2
b2
cP
d1
Client
Proxy
a3
bP
c2
d2
Cloud 3
aP
b3
c3
d3
Cloud 4

(b)

# Cloud interoperability: the Intercloud

- The concern that users could become hopelessly dependent on a single cloud service provider, the so-called vendor lock-in
- An Intercloud ➔ a federation of clouds that cooperate to provide a better user experience.
  organizations increasingly rely on multiple cloud providers, interoperability challenges arise. The concept of the "Intercloud" seeks to address these challenges by enabling seamless interaction between different cloud environments.
- Not likely at this time:
  - There are no standards for either storage or processing.
  - The clouds are based on different delivery models.
  - The set of services supported by these delivery models is large and open; new services are offered every few months.
  - CSPs (Cloud Service Providers) believe that they have a competitive advantage due to the uniqueness of the added value of their services.
  - Security is a major concern for cloud users and an Intercloud could only create new threats.

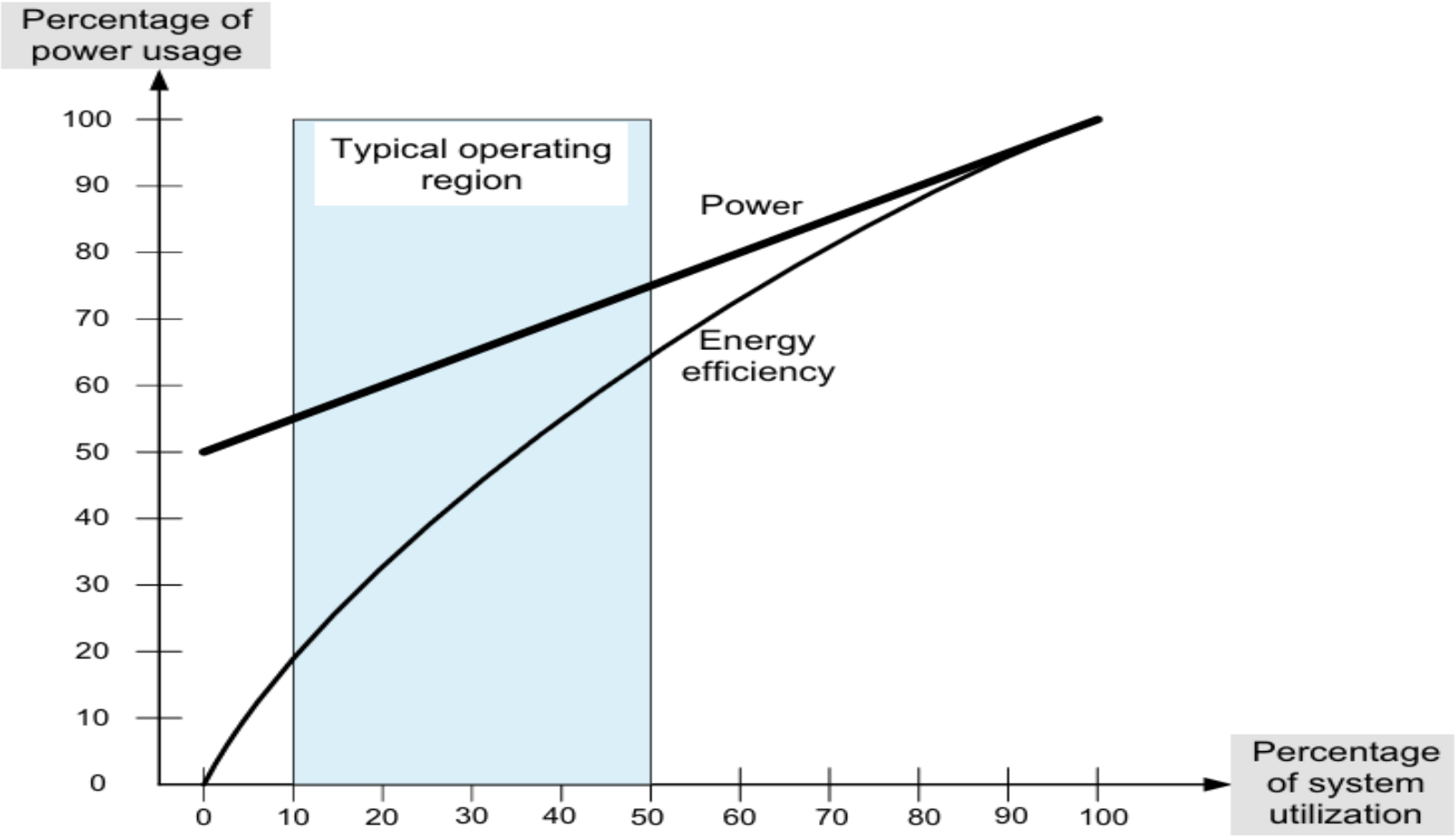
# Energy use and ecological impact

- The energy consumption of large-scale data centers and their costs for energy and for cooling are significant.
- In 2006, the 6,000 data centers in the U.S consumed $61 \times 10^9$ KWh of energy, 1.5% of all electricity consumption, at a cost of $4.5 billion.
- The energy consumed by the data centers was expected to double from 2006 to 2011 and peak instantaneous demand to increase from 7 GWto 12 GW.
- The greenhouse gas emission due to the data centers is estimated to increase from 116 $\times 10^9$ tones of $CO_2$ in 2007 to 257 tones in 2020 due to increased consumer demand.
- The effort to reduce energy use is focused on computing, networking, and storage activities of a data center.
- Operating efficiency of a system is captured by the *performance per Watt of power*.

# Energy use and ecological impact

- Energy-proportional systems strategy for resource management in a computing cloud is to concentrate the load on a subset of servers and switching the rest of the servers to a standby mode whenever possible. This reduce power consumption and, the cost of providing computing and storage services
- In an ideal world, the energy consumed by an idle system should be near zero and should grow linearly with the system load.
- Energy-proportional systems could lead to large savings in energy costs for computing clouds. An energy-proportional system consumes no power when idle, very little power under a light load, and gradually more power as the load increases. example of an energy-proportional network is InfiniBand
- The power consumption required by different types of human activities is partially responsible for the world's greenhouse gas emissions. According to a recent study the greenhouse gas emissions due to data centers are estimated to increase from 150 tons of CO2 in 2007 to 257 tons in 2020, due primarily to increased consumer demand

# Energy use and ecological impact



# Service Level Agreement (SLA)

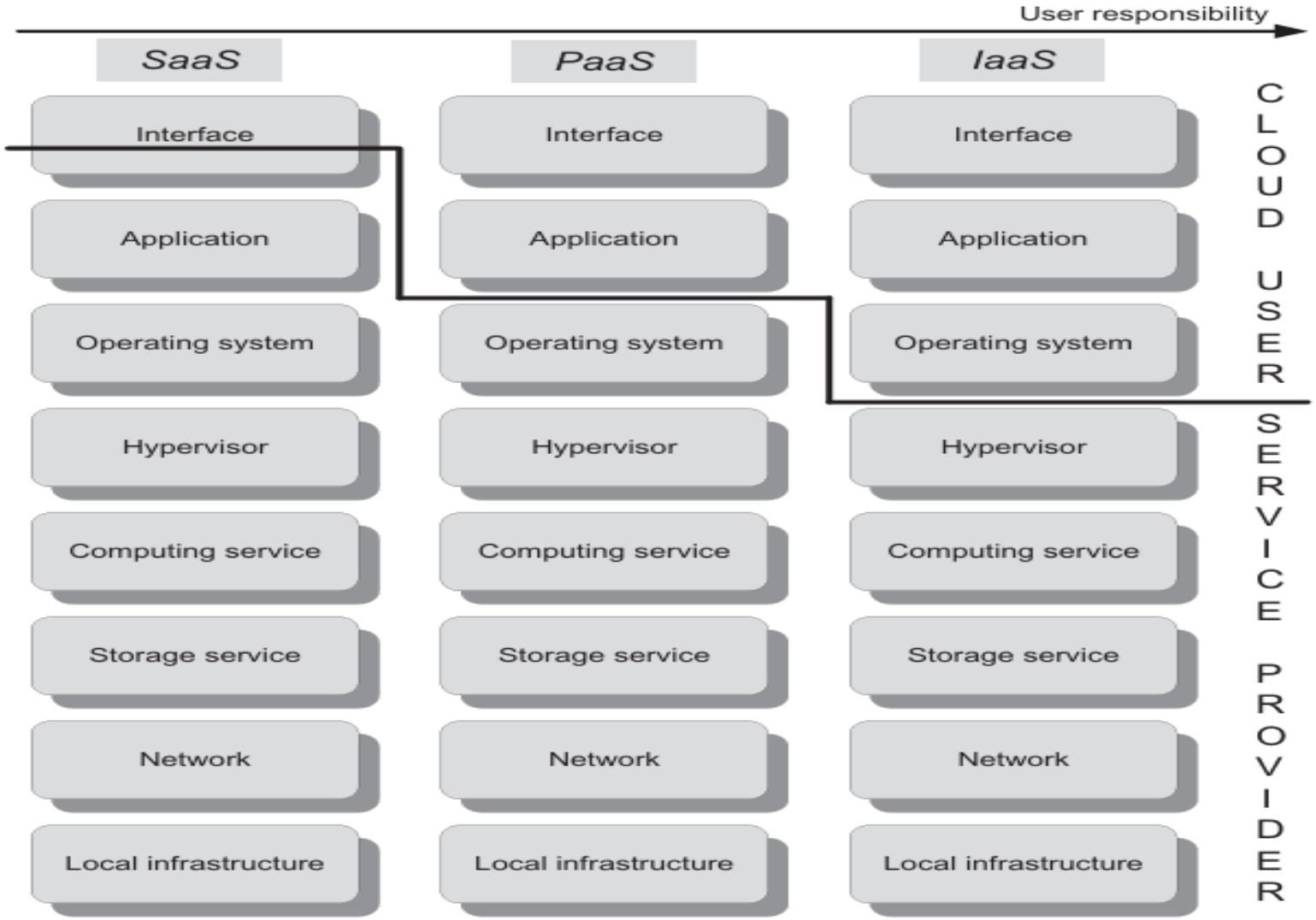
SLA - a negotiated contract between the customer and CSP; can be legally binding or informal.

Objectives:

- Identify and define the customer's needs and constraints including
- Level of resources, security, timing, and QoS.
- Provide a framework for understanding; a critical aspect of this framework is a clear definition of classes of service and the costs.
- Simplify complex issues; clarify the boundaries between the
- Responsibilities of clients and CSP in case of failures.
- Reduce areas of conflict.
- Encourage dialog in the event of disputes.
- Eliminate unrealistic expectations.

- Specifies the services that the customer receives, rather than how the cloud service provider delivers the services.
- An SLA records a common understanding in several areas:
  - (i) services, (ii) priorities, (iii) responsibilities, (iv) guarantees, and (v) warranties.

# Responsibility sharing between user and CSP



# Responsibility sharing between user and CSP

- SaaS service provider supplies both the hardware and the application software, and the user has direct access to these services through a web interface and has no control over cloud resources.
  - Examples of SaaS are Google with Gmail, Google Docs, Google Calendar, Google Groups, and Picasa and Microsoft with the Online Services.
- IaaS, the service provider supplies the hardware (servers, storage, networks) and system software (operating systems, databases); in addition, the provider ensures system attributes such as security, fault tolerance, and load balancing.
  - The representative of IaaS is Amazon AWS.
- PaaS provides only a platform, including the hardware and system software, such as operating systems and databases; the service provider is responsible for system updates, patches, and software maintenance.
- PaaS does not allow any user control of the operating system, security features, or the ability to install applications.
  - Examples are Google App Engine, Microsoft Azure, and Force.com, provided by Salesforce.com.

# Advanced Techniques in Cloud Computing

- **Future Trends in Cloud Computing -** Future trends in cloud computing include increased adoption of multi-cloud and hybrid cloud strategies.
- **Multi-Cloud and Hybrid Cloud Dominance**
  - Hybrid IT Solutions = Cloud + on-premises compute
  - Multi cloud - Presence of more than one cloud service
  - The service is sourced from multiple cloud vendors
  - Distributes workload across multiple cloud environments
  - Ex: A company might use AWS for data storage, Azure for machine learning, and a private cloud for sensitive data.
  - **Hybrid Cloud Computing**

Hybrid cloud = private cloud + public cloud
You can shift your workload between private cloud and public cloud

# Future Trends in Cloud Computing

- **AI and Machine Learning Integration**
  - Cloud platforms will embed AI tools and offer scalable infrastructure for ML workloads.
  - Enables businesses to analyze large datasets, automate tasks, and drive smarter decisions.
  - **Example:** Cloud-native AI services like AWS SageMaker, Google Vertex AI.
- **5G + Cloud Synergy**(cloud services integrated with other technologies like Edge computing)
  - 5G networks enable faster data transmission, enhancing mobile cloud experiences.
  - Powers real-time applications like AR/VR, gaming, and smart cities.
- **Edge Computing Expansion**
  - Data processing is moving closer to the source—especially for real-time applications (IoT, autonomous vehicles).
  - **Example:** AWS Wavelength, Azure Stack Edge, and Google Distributed Cloud.

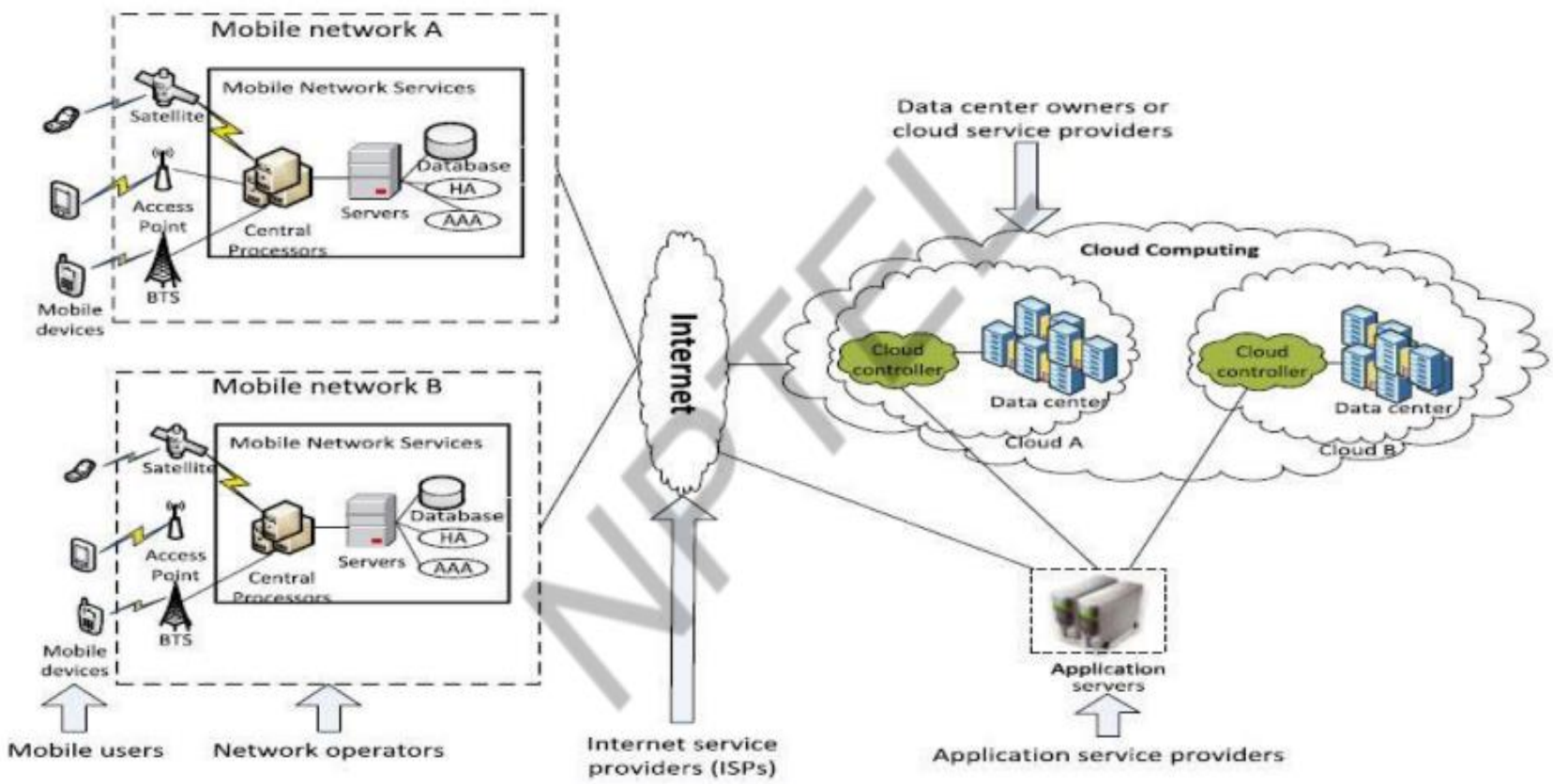
# Mobile Cloud Computing

- A model that combines **mobile computing** and **cloud computing** to deliver rich computing services and applications to mobile users over the internet.
- It allows mobile devices (phones, tablets) to run applications and store data **in the cloud** instead of on the device itself.
- Data storage and data processing happen outside of the mobile device ie into the cloud with low cost and elastically in on-demand fashion
- Mobile device face many resource challenges - like- battery life, storage, bandwidth etc.
- **Examples** Google Photos – Cloud storage and AI-based photo management. Dropbox/OneDrive – File access and sync across mobile and cloud. Spotify – Streaming music from the cloud. Microsoft Office 365 Mobile – Cloud-based document editing.

# Advantages and Disadvantages of MCC

- **Advantages →**
  - Saves battery power
  - Makes execution faster
  - Improves data storage capacity and processing power
  - Improves reliability and availability
  - Dynamic provisioning
- **Disadvantages →**
  - Must send the program states (data) to the cloud server
  - Network latency can lead to execution delay

# Mobile Cloud Computing – Typical Architecture

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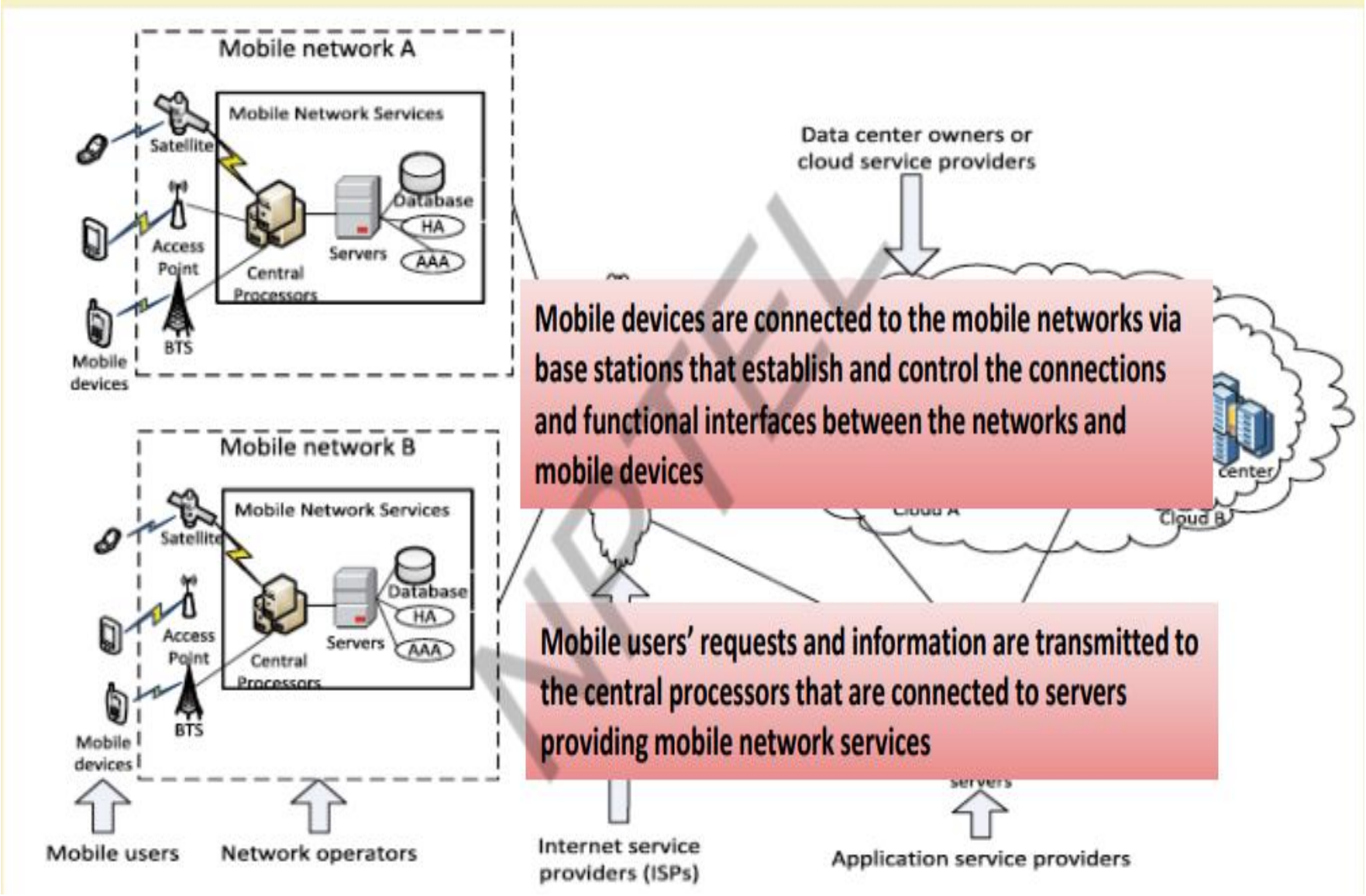
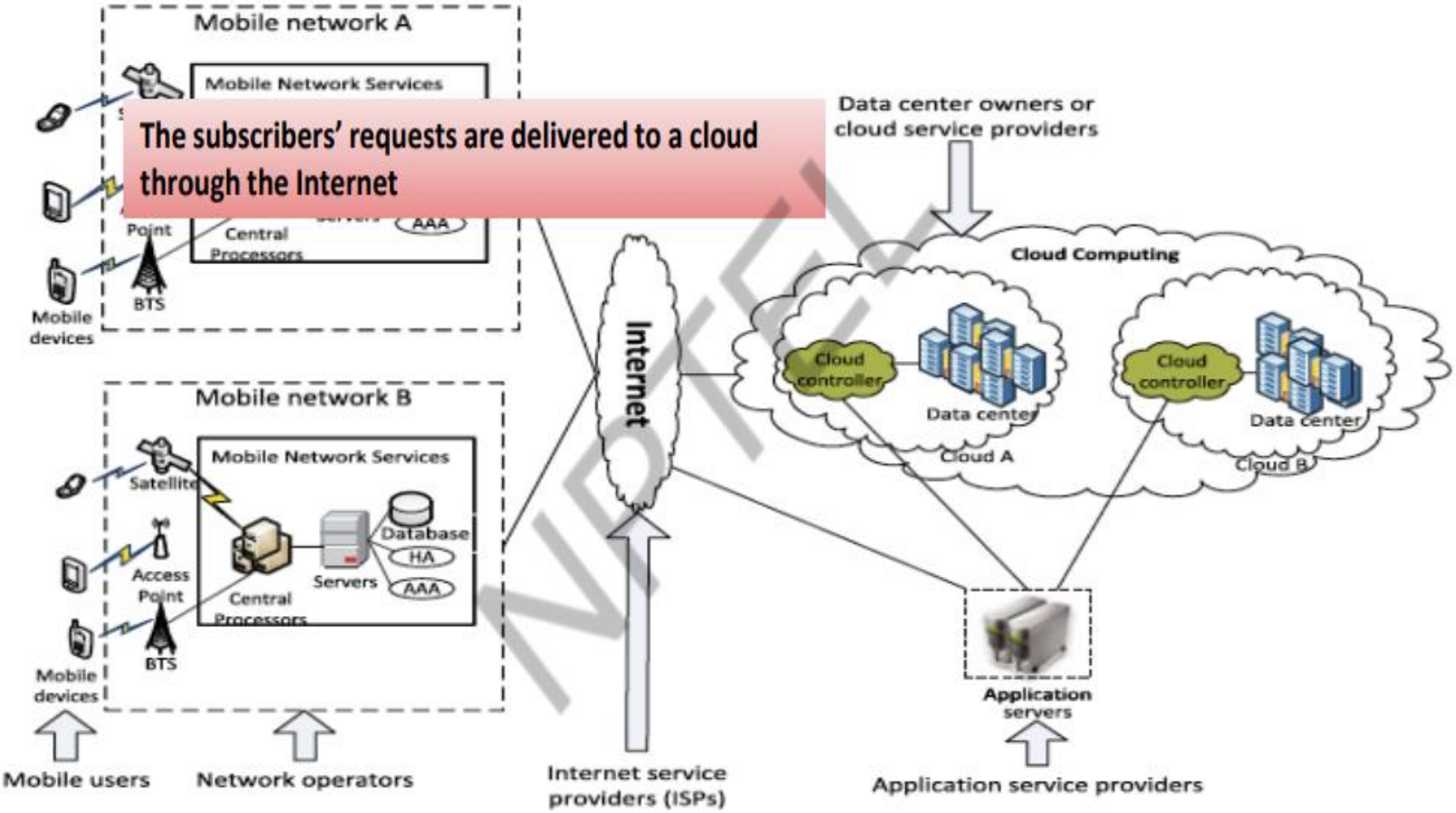



Mobile devices are connected to the mobile networks via base stations that establish and control the connections and functional interfaces between the networks and mobile devices

Mobile users' requests and information are transmitted to the central processors that are connected to servers providing mobile network services

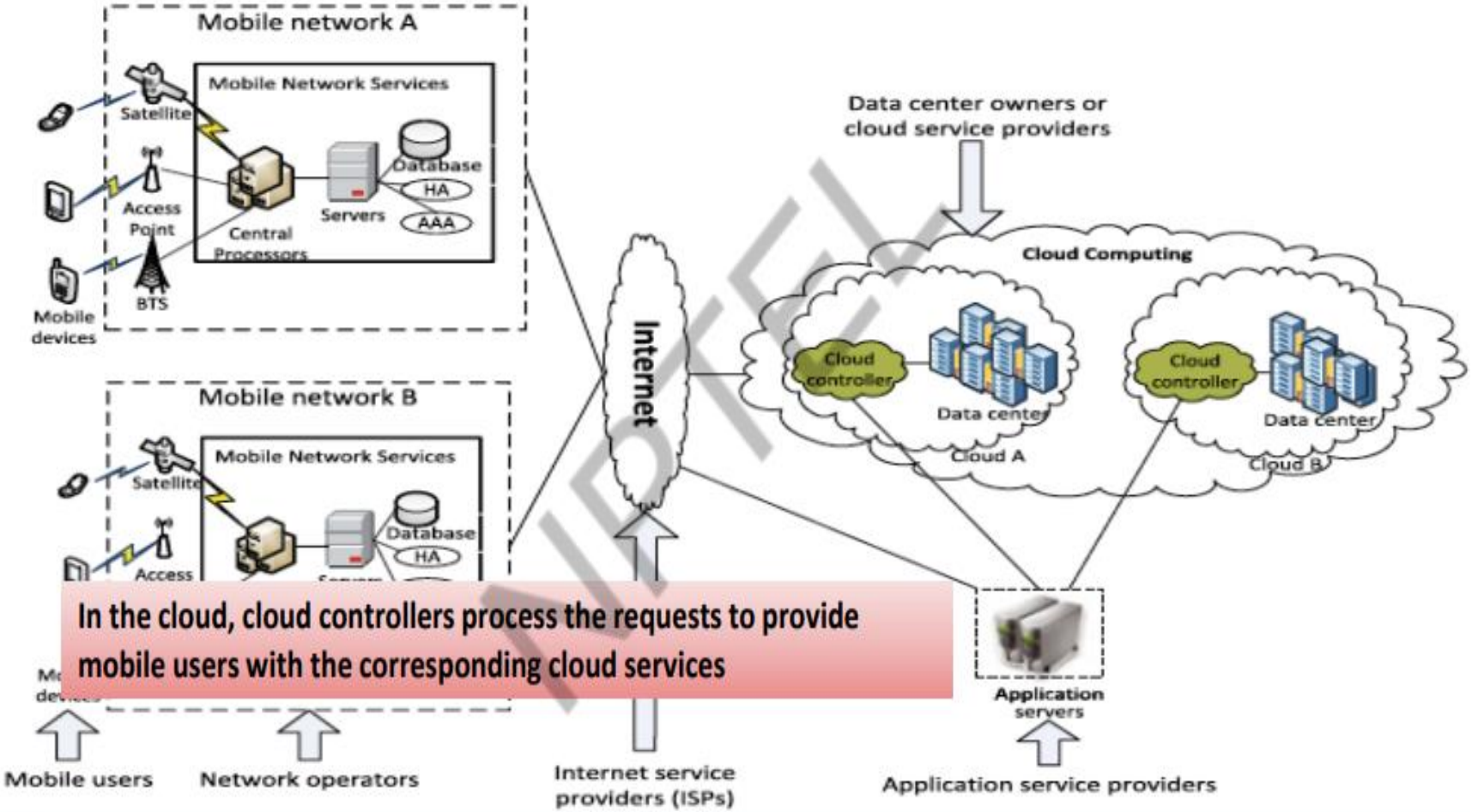
# Mobile Cloud Computing - Architecture




The subscribers' requests are delivered to a cloud through the Internet

NPTEL

# Mobile Cloud Computing - Architecture




In the cloud, cloud controllers process the requests to provide mobile users with the corresponding cloud services

NPTEL

# Mobile Cloud Application

- Mobile Healthcare
  - Mobile healthcare (mHealth) refers to the use of mobile devices (smartphones, tablets, wearables) and wireless technologies to support medical and public health practices. It involves delivering health services, data, and information via mobile technologies.
    - (eg. medical records)
  - M-healthcare offers hospitals and healthcare organizations - a variety of on-demand services on clouds. involves using cloud resources to power mobile applications that provide healthcare services
    - For Ex: Health-Monitoring services, Intelligent emergency management system, Health-aware mobile devices (detect pulse rate, blood pressure, alcohol-level etc.)
  - Mobile Apps – health tracking, consultation, and medication apps. Wearables – smartwatches, fitness bands, ECG monitors. Cloud Computing – secure storage and access to patient data. IoT – Internet-connected medical devices.AI & Big Data – insights from real-time and historical data.

# Mobile Cloud Application

# Mobile Cloud

- **Mobile Learning**
  - Mobile learning (m-learning) refers to learning activities conducted using mobile devices like smartphones and tablets. It enables learning on the go, providing access to educational resources anytime, anywhere.

Enhanced communication quality between

- students and teachers

Help learners to access remote resources

- **Mobile Commerce**
  - M-commerce allows business models for commerce using mobile devices
  - Ex. mobile financial, mobile advertising, mobile shopping

# Types of MCommerce

## Types of Mcommerce

**Mcommerce** refers to all commercial transactions that are completed using a mobile device like a **smartphone or tablet**, including:

| Mobile Shopping | Mobile Banking | Mobile Payments |
| --- | --- | --- |
| App & social commerce, digital content purchases | Mobile money transfers and bill payments | P2P transactions, contactless payments and in-app payments |

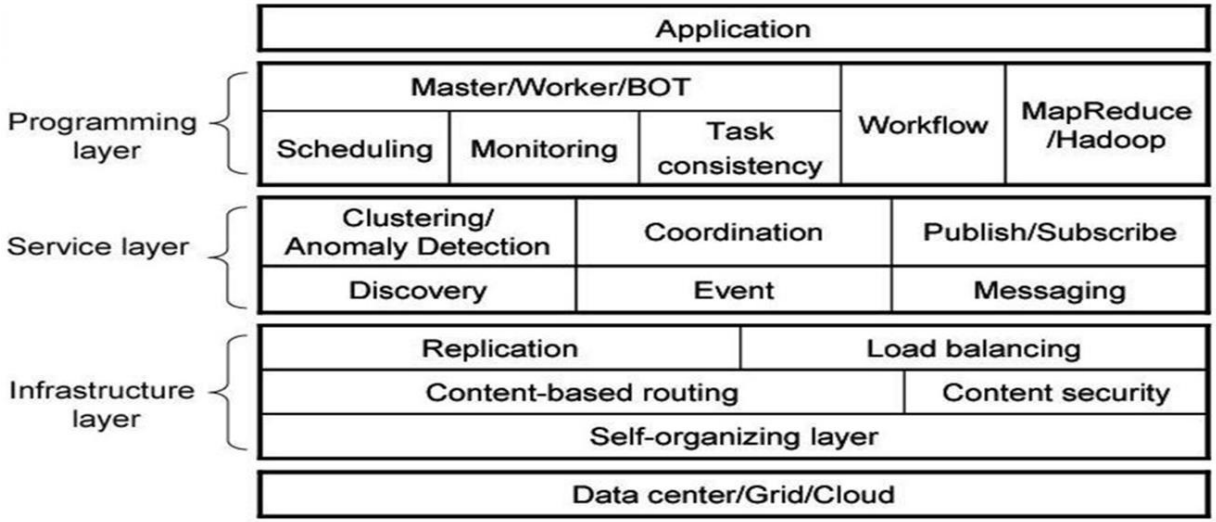
# Autonomic Cloud Computing

- **Autonomic Cloud Computing** refers to cloud systems designed to manage themselves with minimal human intervention by using self-managing features inspired by the human autonomic nervous system.
- These systems automatically handle tasks such as:

| Characteristic | Description |
|---|---|
| **Self-Configuration** | Automatically sets up and configures cloud resources and services. |
| **Self-Optimization** | Continuously monitors and improves system performance and resource usage. |
| **Self-Healing** | Detects faults and recovers from failures automatically. |
| **Self-Protection** | Monitors security threats and takes protective actions without manual input. |

# Comet Cloud - an autonomic cloud engine

- CometCloud allows organizations to build flexible, on-demand cloud systems that can span across multiple cloud providers, grids, data centers, and edge environments — automatically managing resources and workloads. Ex: High-throughput computing, Real-time data analytics, Federated research infrastructures
- **Supports**
  - highly heterogeneous and dynamic cloud infrastructures,
  - integration of public/private clouds, and cloudbursts



# Comet cloud layered abstraction

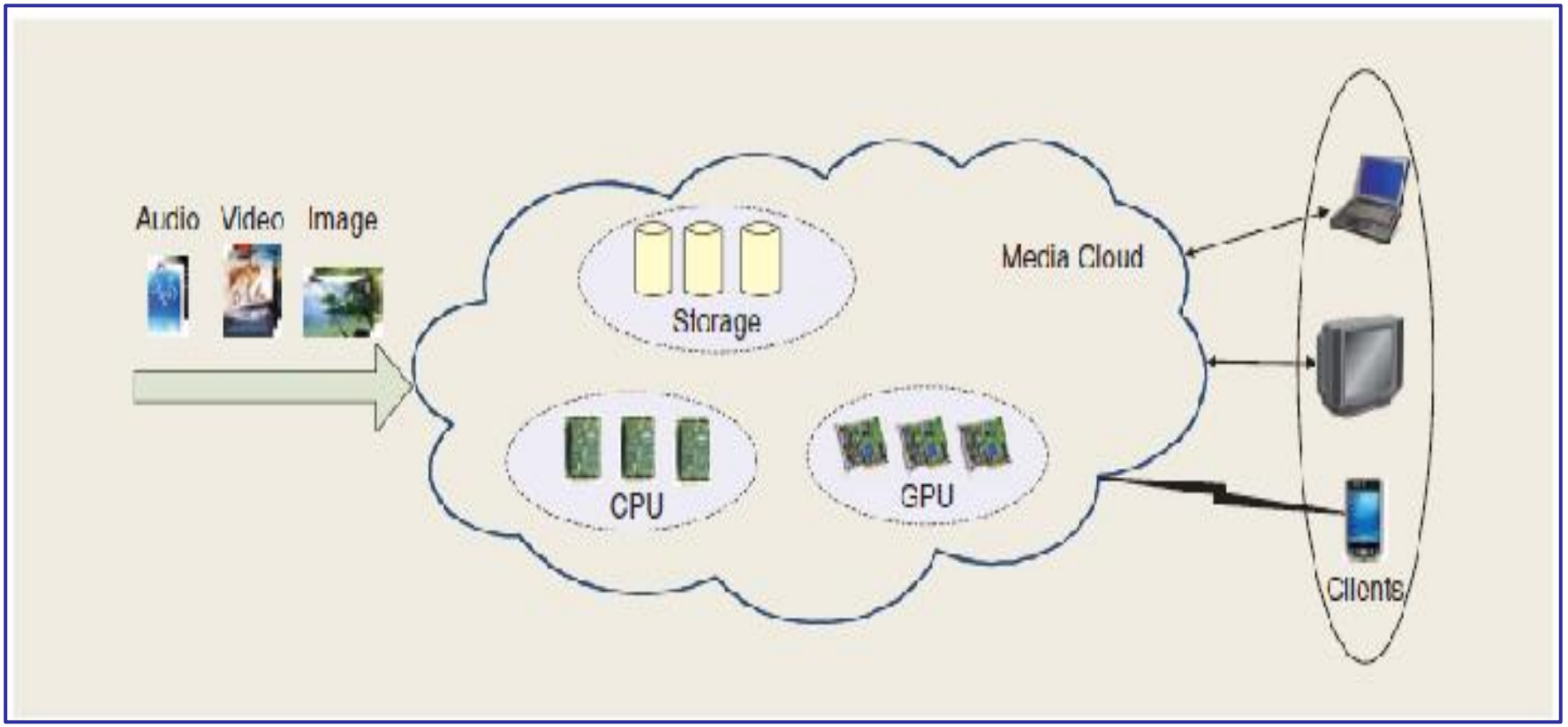
- **Comet is composed of three layers –**
  - Programming Layer
  - Service Layer
  - Infrastructure Layer
- **Programming Layer** provides -basic framework for application, development and management
  - It supports range of paradigms like - master / worker
  - Masters generate tasks and workers consume them

**The service layer** -provides a range of services -

- Handles scheduling, resource management, and autonomic behavior
- **The infrastructure layer** Provides - Load Balancing, Data Replication, Content based routing, Self organizing layer. Federates resources from private clouds, public clouds (AWS, Azure), grids, and data centers.

# Multimedia Cloud

- Multimedia Cloud refers to cloud computing services specially designed to store, process, and deliver multimedia content—such as videos, images, audio, and interactive media—over the internet.



# Multimedia Cloud

| Service | Function |
|---|---|
| **YouTube** | Video hosting and streaming |
| **Netflix** | Cloud-based video streaming |
| **Adobe Creative Cloud** | Cloud media editing and storage |
| **Twitch** | Live video game streaming |
| **Spotify** | Cloud music streaming |

**IPTV (Internet Protocol Television)** is a technology that delivers television content over Internet Protocol (IP) networks instead of traditional terrestrial, satellite, or cable formats. You can watch your favorite TV shows from computers, laptops, and mobile devices.
Ex: Smart TV, Apple TV, Google chromecast,

# Multimedia Cloud

- **OTT** media service is a media service offered directly to viewers via the Internet. OTT bypasses cable, broadcast, and satellite television platforms.
  - Ex: Direct-to-Consumer services With Low Prices, Advanced User Interfaces and Easy-to-Access Multi Stream Video
- **OTT messaging** is defined as instant messaging services or onlinechat provided by third parties as an alternative to text messaging services provided by a mobile network operator.
  - For Ex. WhatsApp, Viber, WeChat, iMessage, Skype, Telegram.
- **OTT voice calling** refers to **voice communication services** delivered via the **internet.** as provided by FaceTime, Skype, Viber, WhatsApp, WeChat, and Zoom

# Multimedia Cloud

| | OTT (Over-the-top) | IPTV (Internet Protocol TV) |
|---|---|---|
| Content Delivery | Uses open internet, unmanaged network "open ecosystem" | Uses dedicated, managed network. " Walled Garden ecosystem" |
| Network Type | Delivered from the provider/content aggregator to the viewer using open network | Closed, proprietary network, accessed via a specific internet service provider |
| Network Ownership | Without the need for intervening carriage negotiations, or infrastructure investments | Services are optimized and customized to suit the network and end-device capability |
| Quality of Service (QOS ) | Not guaranteed, works under best effort conditions | Enables control over quality of delivery |
| Examples | Popular Video on demand services like YouTube, Netflix, Amazon Lovefilm, BBC iPlayer, Hulu etc. | Service Example includes U-verse (AT&T), PrismTv (Century Link) |
| Protocol | Delivered using HTTP (TCP), a connected transport protocol. Emerging trends using adaptive streaming technologies like HLS (Apple) , Smooth Streaming(Microsoft) and HDS(Adobe) | Traditional IPTV uses TS (transport stream) transmission technology. Delivers content over UDP, connectionless protocol |
| Content Catalog | Widely used for freemium and economical VOD delivery models | Used primarily for premium content and real time content delivery like broadcasting TV |
| Content Type | Generally not premium due to security/DRM Concerns | Premium Content |
| Routing Topology | Unicast (Based on HTTP) | Multicast, Unicast burst during channel change leading multicast join |
| Category | Complementary Service | Parallel service category to Cable/Satellite |
| Major Players | OVP (Kaltura, Brightcove), CDN Players (Akamai, L3, Limelight) and Content Aggregators | TSP and IPTV Platform vendors – Microsoft Mediaroom (Ericsson), ALU, Cisco.. |
| Key Challenges | Low Quality, Non Premium Content<br>No Live Broadcast, Unicast model | Expensive, Competition from Cable/DTH industry, Bandwidth and Infrastructure |
| Key Benefits | Low cost, Flexibility of content consumption across devices | Interactive Service, Quality of Service and Quality of Experience |

# Energy Aware Cloud Computing

- Green Computing (or Green IT) is the practice of environmentally sustainable computing that focuses on reducing the energy consumption, e-waste, and overall environmental impact of IT systems. It applies to hardware, software, data centers, and user practices.
- **Key Issues**
  - High Power Consumption
  - Cooling Requirements
  - Underutilization of Resources
  - Carbon Footprint
  - Lack of Energy-Aware
  - Security vs. Energy Trade-offs
  - Data Transfer Energy
  - Infrequent Hardware Refresh

**Best Practices**:

- Use green-certified data centers (e.g., LEED-certified)
- Adopt renewable energy sources (solar, wind)
- Implement auto-scaling policies
- Use serverless computing to optimize resource usage
- Encourage geographical load balancing (use data centers in cooler climates or near renewable sources)
- Energy-Efficient Data Centers

# Green Computing

**Green use**

Minimizing the electricity consumption of computers and their peripherals by Using them in eco-friendly manner

**Green disposal**

Reusing old devices or recycling or disposing them in proper way

**Green design**

Designing energy efficient computers, servers, printers, projectors and other digital devices

**Green manufacturing**

Minimizing waste during manufacturing of computers and other subsystems

# Jungle Computing

- Jungle Computing refers to a complex, heterogeneous, and highly distributed computing environment that combines various types of computing resources such as: Clouds (public/private), clusters, grids, desktop grids, supercomputers, stand alone machines and mobile devices.
- To efficiently run complex, large-scale, or parallel scientific and technical applications using whatever computing resources are available—regardless of location, type, or ownership.

# Jungle Computing

- Examples:

A scientific research project might use a cloud platform for data storage, a local cluster for intensive calculations, and a grid system for simulations, all as part of a jungle computing environment.